INSPECTION REPORT

# ChArUco target · ceramic · 185 × 110 mm

ITEM

| | |
|---|---|
| Part number | CT-CHA-C-185110-6X6 |
| Description | ChArUco calibration target, ceramic, 185 × 110 mm |
| Serial number | 000311 |
| Lot | L-2611-0382 |
| Datasheet | CT-DS-CHA-C · Rev A |
| Pattern | ChArUco 7 × 12 · 14 / 10.5 mm DICT_6X6_250 |
| Marker ID range | 0–41 (42 markers) |
| Start ID | 0 |

INSPECTION

| | |
|---|---|
| Inspection date | 2026-10-24 |
| Inspector | JG |
| Location | per Calib.Tools procedure CT-WI-001 |
| Instrument | Optical CMM · OCMM-01 |
| Last calibration | 2026-03 · valid to 2027-03 |
| Temperature | 20.4 °C ambient · 20 °C reference |

OVERALL RESULT

**PASS**

All measured features within datasheet tolerance.

9 geometry · 1 flatness · 4 visual
66 corners · U = 1.5 µm (k = 2)

## 1 Method

All 66 ChArUco corners were measured by comparison against golden sample GS-CHA-C-185110-6X6 (DAkkS-calibrated, certificate [number]) on Optical CMM (OCMM-01); flatness was measured at 25 points. Expanded uncertainty U = 1.5 µm (k = 2), combining golden-sample certificate and comparison uncertainties. Results corrected to 20 °C reference; ambient 20.4 °C.

## 2 Pattern geometry

ceramic: deviations in µm

| FEATURE | NOMINAL (mm) | MEASURED (mm) | DEVIATION (µm) | TOLERANCE (µm) | RESULT |
|---|---|---|---|---|---|
| Square pitch X | 14.000 | 14.0008 | +0.8 | ±2 | PASS |
| Square pitch Y | 14.000 | 13.9994 | −0.6 | ±2 | PASS |
| Pattern width | 168.000 | 168.003 | +3 | ±5 | PASS |
| Pattern height | 98.000 | 98.002 | +2 | ±5 | PASS |
| Diagonal 1 | 194.454 | 194.457 | +3 | ±7 | PASS |
| Diagonal 2 | 194.454 | 194.452 | −2 | ±7 | PASS |
| Orthogonality | 90.000° | 90.0006° | – | ±0.002° | PASS |
| Max single-feature deviation · n = 66 | – | – | 1.4 | ≤ 2 | PASS |
| RMS deviation · information only | – | – | 0.6 | – | N/A |

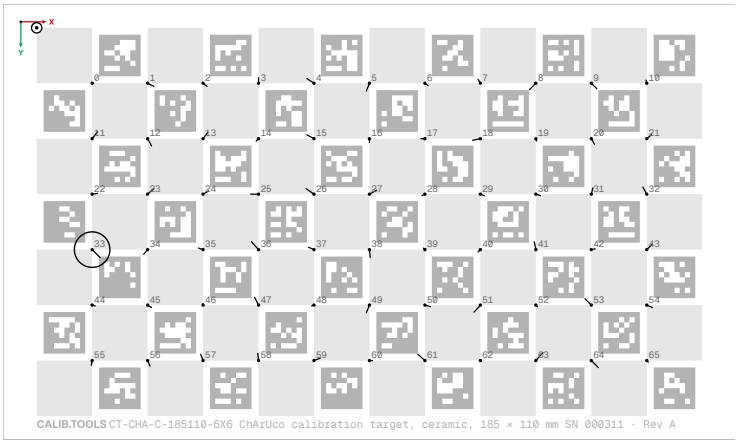
## 3 Flatness

| MEASURED FLATNESS (µm) | TOLERANCE (µm) | METHOD | RESULT |
|---|---|---|---|
| 12 | ≤ 20 | 25 points, 5 × 5 grid, least-squares plane | PASS |

## 4 Visual

| CHECK | CRITERION | RESULT |
|---|---|---|
| Edge chamfer | 0.5 × 45° on all edges, no chips > 0.3 mm | PASS |
| Surface defects | No scratches or pinholes in the pattern area | PASS |
| Print / edge definition | Square edges sharp, no ink bleed at 10× loupe | PASS |
| Marking legibility | Label line, X / Y arrows and SN legible | PASS |

## 5 Feature map




LEGEND

X → right

Y ↓ down

⊙ Origin (0, 0): top-left outer corner of the pattern (OpenCV frame)

• Measured corner with id, n = 66

•—— Deviation vector, magnified × 2000 (1 µm = 2 mm on the board)

◯ Largest deviation: id 33, 1.4 µm

## 6 Measured coordinates

Complete corner coordinates are supplied in CT-CSV-000311.csv (66 rows; columns id, row, col, x_nom, y_nom, x_meas, y_meas, dx_um, dy_um; mm unless stated). First 10 rows:

| id | row | col | x_nom | y_nom | x_meas | y_meas | dx_um | dy_um |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 14.000 | 14.000 | 14.0002 | 14.0000 | +0.2 | 0.0 |
| 1 | 0 | 1 | 28.000 | 14.000 | 28.0008 | 14.0004 | +0.8 | +0.4 |
| 2 | 0 | 2 | 42.000 | 14.000 | 42.0005 | 14.0004 | +0.5 | +0.4 |
| 3 | 0 | 3 | 56.000 | 14.000 | 56.0000 | 13.9993 | 0.0 | −0.7 |
| 4 | 0 | 4 | 70.000 | 14.000 | 69.9991 | 13.9994 | −0.9 | −0.6 |
| 5 | 0 | 5 | 84.000 | 14.000 | 83.9996 | 14.0010 | −0.4 | +1.0 |
| 6 | 0 | 6 | 98.000 | 14.000 | 98.0004 | 14.0003 | +0.4 | +0.3 |
| 7 | 0 | 7 | 112.000 | 14.000 | 111.9997 | 13.9995 | −0.3 | −0.5 |
| 8 | 0 | 8 | 126.000 | 14.000 | 125.9992 | 14.0008 | −0.8 | +0.8 |
| 9 | 0 | 9 | 140.000 | 14.000 | 140.0007 | 14.0007 | +0.7 | +0.7 |

## 7 Statement

The item identified above meets the specifications of datasheet CT-DS-CHA-C rev A at the time of inspection. This is an inspection report issued by Calib.Tools; it is not an accredited calibration certificate. An ILAC-MRA certificate issued by an accredited third-party laboratory is available as SVC-X-CERT-ILAC.

## 8 Traceability

| | |
|---|---|
| Instrument | OCMM-01 · certificate K-2026-03-0141 · valid to 2027-03 |
| Reference standard | Golden sample GS-CHA-C-185110-6X6 · DAkkS certificate [number] · valid to [date] |
| Raw data | Raw data retained for 10 years under this serial number. |

SIGNED

JG

JG · Inspector · 2026-10-24
Printed signature

CALIB.TOOLS SN 000311 INSPECTED 2026-10-24 JG DIMENSIONAL INSPECTION

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